ISE 218: Fundamentals of Information Technology

Practice Problems for Unit 3: Basic Computer Architecture

Directions: Below is a set of exercises and problems you should do for practice. Answering this set of questions should be just one part of your preparation for quizzes and exams. In addition to solving these problems you should:

- Review your homework assignments.
- Review the lecture notes and read the relevant sections of the course textbook.
- Do additional practice problems from the textbook; see the problems at the end of each chapter.
- Form a study group and test each other on the material.

1. How many bits are required to address a 1M × 8 main memory if
   a. main memory is byte-addressable?

      There are 1M × 1 bytes, or 2^{20} total bytes, so 20 bits are needed for an address

   b. mMain memory is word-addressable?

      There are 1M words, or 2^{20} total words, so 20 bits are required for an address

2. How many bits are required to address a 2M × 32 main memory if:
   a. main memory is byte-addressable?

      The memory is $2\times2^{20}\times4=2^{1}\times2^{20}\times2^{2}=2^{23}$ bytes in size. If the memory is byte-addressable, 23 bits are therefore needed for a memory address.

   b. main memory is word-addressable?

      There are $2\times2^{20}=2^{21}$ words of memory. If the memory is word-addressable, 21 bits are needed for a memory address.

3. How many bits are required to address a 16K × 64 main memory if:
   a. main memory is byte-addressable?

      The memory is $16\times2^{10}\times8=2^{4}\times2^{10}\times2^{3}=2^{17}$ bytes in size. If the memory is byte-addressable, 17 bits are therefore needed for a memory address.

   b. main memory is word-addressable?

      There are $16\times2^{10}=2^{14}$ words of memory. If the memory is word-addressable, 14 bits are needed for a memory address.

4. A 4KB memory is built using 256 × 8 RAM chips.

a. How many bits will each address contain?

12 ($4096=2^{12}$)

b. How many lines must go to each chip?

8 ($256=2^8$)

c. How many chips do we need?

The memory is 4096 bytes in size and composed of 256 byte chips. $4096/256 = 2^{12}/2^8 = 2^4 = 16$ chips are needed.

d. How many lines (bits) must be decoded for the *chip select* inputs?

We have 16 chips so we need 4 lines (bits) to select the chip.

5. Suppose that a 1M × 32 main memory is built using 256K × 16 RAM chips and memory is word-addressable.

a. How many RAM chips are necessary?

The memory is $1\times2^{20}\times4=1\times2^{20}\times2^2=2^{22}$ bytes in size.

Each RAM chip is $256\times2^{10}\times2=2^8\times2^{10}\times2=2^{19}$ bytes in size.

Therefore $2^{22}/2^{19}=2^3=8$ RAM chips are needed.

b. How many address bits are needed to address one memory cell in a RAM chip?

Since each RAM chip has $2^8\times2^{10}=2^{18}$ memory cells, 18 bits are needed to address one 16-bit memory cell in a chip.

c. How many address bits are needed for all of memory?

The memory is 2^{20} words in size, and memory is word-addressable. Therefore, 20 bits are needed to address the memory.

6. Suppose that a 4M × 16 main memory is built using 128K × 8 RAM chips and memory is word-addressable.

a. How many RAM chips are necessary?

The memory is $4\times2^{20}\times2=2^2\times2^{20}\times2^1=2^{23}$ bytes in size.

Each RAM chip is $128\times2^{10}\times1=2^7\times2^{10}=2^{17}$ bytes in size.

Therefore $2^{23}/2^{17}=2^6=64$ RAM chips are needed.

b. How many address bits are needed to address one byte in a RAM chip?

Since each RAM chip has 2^{17} bytes, 17 bits are needed to address one byte in a chip.

c. How many address bits are needed for all of memory?

The memory is 2^{22} words in size, and memory is word-addressable. Therefore, 22 bits are needed to address the memory.

7. Suppose we have a 2^{18}-byte memory.

   a. What are the lowest and highest addresses if memory is byte-addressable?

      There are 2^{18} bytes, which can all be addressed using addresses 0 through $2^{18}-1$.

   b. What are the lowest and highest addresses if memory is word-addressable, assuming a 16-bit word?

      There are only 2^{17} words and addressing each requires using addresses 0 through $2^{17}-1$.

   c. What are the lowest and highest addresses if memory is word-addressable, assuming a 64-bit word?

      There are only 2^{15} words and addressing each requires using addresses 0 through $2^{15}-1$.

8. Given the instruction set for MARIE, decode the following MARIE machine language instructions (write the assembly language equivalent). Write any numerical values in hexadecimal. (Refer to the table marked "MARIE Instruction Set" at this end of this homework assignment to help you solve this problem.)

   a. 1010000000000000

      Clear

   b. 0011 111111101010

      Add FEA

   c. 0111000000000000

      Halt

   d. 0010000001000111

      Store 47

9. Consider the instruction LOAD 10D. Assume that the value 0x0014 is stored at memory location 0x010D. Use *register transfer notation* to show how the values of PC, IR, MAR, MBR and AC change with the execution of each micro-operation for this particular instruction. Your table should resemble the ones from the lecture notes:

Step	RTN	PC	IR	MAR	MBR	AC

   Use as many rows as needed.

   Use the following as the starting values of the five registers (all are hexadecimal values):

   - PC: 108
   - IR: 9108
   - MAR: 108
   - MBR: 000C

- AC: FFF8

(Refer to the lecture notes and the table marked "MARIE Instruction Set" at this end of this document to help you solve this problem.)

Solution:

Step	RTN	PC	IR	MAR	MBR	AC
(initial values)		108	9108	108	000C	FFF8
Fetch	MAR ← PC	108	9108	108	000C	FFF8
	IR ← M[MAR]	108	110D	108	000C	FFF8
	PC ← PC + 1	109	110D	108	000C	FFF8
Decode	MAR ← IR[11-0]	109	110D	10D	000C	FFF8
	(decode IR[15-12]	109	110D	10D	000C	FFF8
Get operand	MBR ← M[MAR]	109	110D	10D	0014	FFF8
Execute	AC ← MBR	109	110D	10D	0014	0014

10. Consider the instruction ADD 10B. Assume that the hexadecimal value 0030 is at memory location 0x10B. Use *register transfer notation* to show how the values of PC, IR, MAR, MBR and AC change with the execution of each micro-operation for this particular instruction. Your table should resemble the ones from the lecture notes:

Step	RTN	PC	IR	MAR	MBR	AC

Use as many rows as needed.

Use the following as the starting values of the five registers (all are hexadecimal values):

- PC: 10E
- IR: 110B
- MAR: 10B
- MBR: 0030
- AC: 0030

Solution:

Step	RTN	PC	IR	MAR	MBR	AC
(initial values)		10E	110B	10B	0030	0030
Fetch	MAR ← PC	10E	110B	10E	0030	0030
	IR ← M[MAR]	10E	310B	10E	0030	0030
	PC ← PC + 1	10F	310B	10E	0030	0030
Decode	MAR ← IR[11-0]	10F	310B	10B	0030	0030
	(decode IR[15-12]	10F	310B	10B	0030	0030
Get operand	MBR ← M[MAR]	10F	310B	10B	0030	0030
Execute	AC ← AC+ MBR	10F	310B	10B	0030	0060

11. Consider the MARIE instruction Store 4FD, which stores the contents of the accumulator at hexadecimal memory address 0x04FD. Assume the following as the starting values of the five registers (all are hexadecimal values):

- PC: E2F
- IR: 512C
- MAR: 115
- MBR: 003A
- AC: 0046

Fill in the table below to show how the values of PC, IR, MAR, MBR and AC change with the execution of each micro-operation. Refer to the last page of the exam to help you solve this problem.

Step	**RTN**	**PC**	**IR**	**MAR**	**MBR**	**AC**
(init. values)		E2F	512C	115	003A	0046
Fetch	MAR <- PC	E2F	512C	**E2F**	003A	0046
	IR <- M[MAR]	E2F	**24FD**	E2F	003A	0046
	PC <- PC + 1	**E30**	24FD	E2F	003A	0046
Decode	MAR <- IR[11:0]	E30	24FD	**4FD**	003A	0046
	(Decode IR[15:12])	E30	24FD	4FD	003A	0046
Get Operand	(not necessary)	E30	24FD	4FD	003A	0046
Execute	MBR <- AC	E30	24FD	4FD	**0046**	0046
	M[MAR] <- MBR	E30	24FD	4FD	0046	0046

12. Consider the MARIE instruction Load F4B, which copies the 16-bit value stored at memory address 0x0F4B into the accumulator. Suppose the hexadecimal value 3AE2 is stored at memory address 0x0F4B. Assume the following as the starting values of the five registers (all are hexadecimal values):

- PC: 2AF
- IR: F92E
- MAR: 71C
- MBR: 00C2
- AC: 0081

Fill in the table below to show how the values of PC, IR, MAR, MBR and AC change with the execution of each micro-operation. Refer to the last page of the exam to help you solve this problem.

Step	**RTN**	**PC**	**IR**	**MAR**	**MBR**	**AC**
(init. values)		2AF	F92E	71C	00C2	0081
Fetch	MAR <- PC	2AF	F92E	**2AF**	00C2	0081

	IR <- M[MAR]	2AF	**1F4B**	2AF	00C2	0081
	PC <- PC + 1	**2B0**	1F4B	2AF	00C2	0081
Decode	MAR <- IR[11:0]	2B0	1F4B	**F4B**	00C2	0081
	(Decode IR[15:12])	2B0	1F4B	F4B	00C2	0081
Get Operand	MBR <- M[MAR]	2B0	1F4B	F4B	**3AE2**	0081
Execute	AC <- MBR	2B0	1F4B	F4B	3AE2	**3AE2**

MARIE Instruction Set

Opcode	Instruction	RTN
0000	JnS *X*	MBR ← PC MAR ← *X* M[MAR] ← MBR MBR ← *X* AC ← 1 AC ← AC + MBR PC ← AC
0001	Load *X*	MAR ← *X* MBR ← M[MAR] AC ← MBR
0010	Store *X*	MAR ← *X*, MBR ← AC M[MAR] ← MBR
0011	Add *X*	MAR ← *X* MBR ← M[MAR] AC ← AC + MBR
0100	Subt *X*	MAR ← *X* MBR ← M[MAR] AC ← AC - MBR
0101	Input	AC ← InREG
0110	Output	OutREG ← AC
0111	Halt	
1000	Skipcond	If IR[11–10] = 00 then If AC < 0 then PC ← PC + 1 Else If IR[11-10] = 01 then If AC = 0 then PC ← PC + 1 Else If IR[11-10] = 10 then If AC > 0 then PC ← PC + 1
1001	Jump *X*	PC ← IR[11-0]
1010	Clear	AC ← 0